

Problem Set: Magnetic Methods

Geophysics 3A

Question 1: Magnetic Anomalies of Simple Bodies

[10 pts.]

This question builds on **Week 5, Activity 2: Magnetic Structures**.

For each of the following bodies assume:

- a uniform vertical inducing geomagnetic field with,
- constant magnetic susceptibility within each body,
- magnetization is purely induced (no remanent magnetization),
- the observed quantity is the vertical magnetic anomaly, B_z .

(A) Conceptual predictions

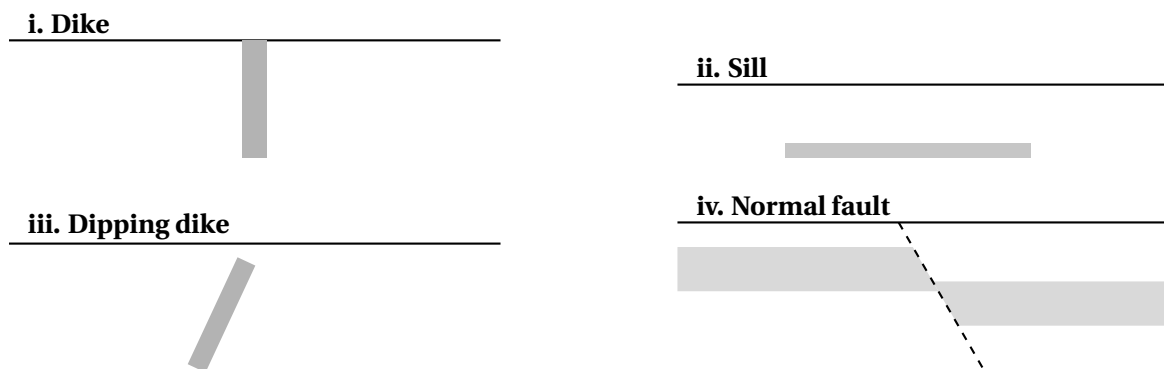


Figure 1: Idealized 2-D susceptibility structures used to explore magnetic anomaly geometry.

For each geometry:

1. Sketch the expected vertical magnetic anomaly as a function of horizontal distance.
2. Indicate the locations of maxima and minima relative to the body.
3. Comment on symmetry or asymmetry of the anomaly.
4. Describe the far-field (long-wavelength) behavior.
5. Compare qualitatively with the gravity anomaly for the same structure.

(B) Forward modelling

Using the Talwani magnetic modelling code (`gravmag2d_gui.py`):

1. Construct polygons corresponding to the bodies above.
2. Assign a constant magnetic susceptibility contrast.
3. Compute the vertical magnetic anomaly $B_z(x)$.
4. Compare modelled anomalies to your sketches.
5. For the vertical dike, investigate how burial depth, susceptibility contrast, body width, and field inclination affect anomaly wavelength and amplitude.

Question 2: Dipole Anomalies

[10 pts.]

For a vertical magnetic dipole ($\hat{m} = \hat{z}$) the radial and azimuthal components of the magnetic field in spherical coordinates (from the notes) are

$$\begin{aligned} B_r &= \frac{\mu_0}{2\pi} \frac{m \cos \theta}{r^3}, \\ B_\theta &= \frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^3}, \\ B_\phi &= 0, \end{aligned}$$

where r and θ are the distance to and the co-latitude of the dipole at a point P . However, since we wish to know the vertical component of the field, we must convert the field to Cartesian coordinates from spherical. The conversion of a vector in spherical, $B(r, \phi, \theta)$, to Cartesian coordinates, $B(x, y, z)$, is performed by

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix},$$

where

$$\begin{aligned} r &= (x^2 + y^2 + z^2)^{1/2} \\ \phi &= \tan^{-1} \frac{y}{x} \\ \theta &= \cos^{-1} \frac{z}{r} = \sin^{-1} \frac{(x^2 + y^2)^{1/2}}{r}. \end{aligned}$$

Note that in for these problems, we can assume $y = 0$ and $\phi = 0$.

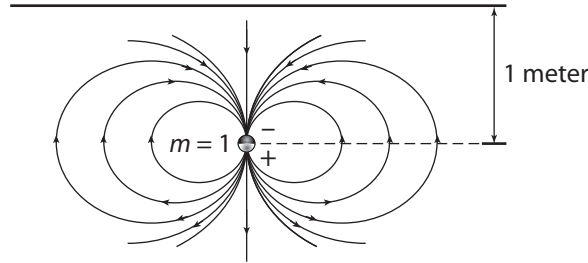


Figure 2: Vertical magnetic dipole with moment $m = 1$ at 1 m below the surface.

1. Show that the vertical magnetic field due to a vertically oriented dipole is given by,

$$B_z = \frac{\mu_0}{4\pi} \frac{m}{r^3} [3 \cos^2 \theta - 1]. \quad (1)$$

Remember, $\cos^2 \theta + \sin^2 \theta = 1$.

2. If we wish to know the vertical field at the surface of the Earth due to a buried dipole with an arbitrary rotation of the dipole axis, $\hat{m}$, about $\hat{y}$. We define the rotation angle, α , such that

$$\theta = \theta' + \alpha,$$

where θ is the angle between the $\hat{z}$ and P , θ' is the angle between the $\hat{m}$ and the point P , and α is the

angle between $\hat{z}$ and $\hat{m}$. In this case the magnetic field is given by

$$B_{r'} = \frac{\mu_0}{2\pi} \frac{m \cos \theta'}{r^3},$$

$$B_{\theta'} = \frac{\mu_0}{4\pi} \frac{m \sin \theta'}{r^3}.$$

Note that $r' = r$ since the distance between the dipole and P does not change under rotation. The vertical field for a rotated dipole is

$$B_z = B_{r'} \cos \theta - B_{\theta'} \sin \theta$$

$$= \frac{\mu_0}{4\pi} \frac{m}{r^3} [2 \cos(\theta - \alpha) \cos \theta - \sin(\theta - \alpha) \sin \theta].$$

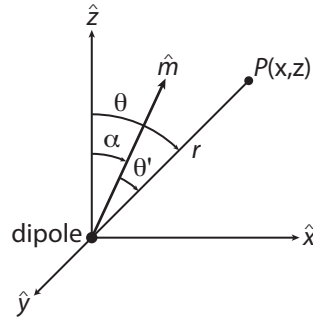


Figure 3: Coordinate system for a dipole with arbitrary rotation of the dipole axis, $\hat{m}$, by an angle α w.r.t. $\hat{z}$.

Show that a horizontal dipole, $\alpha = \frac{\pi}{2}$, the vertical magnetic is therefore given by

$$B_z = \frac{3\mu_0}{4\pi} \frac{m}{r^3} \cos \theta \sin \theta. \quad (2)$$

Recall that $\cos(\theta - \frac{\pi}{2}) = \sin \theta$ and $\sin(\theta - \frac{\pi}{2}) = -\cos \theta$. Note: this result is identical to B_x for a vertical dipole.

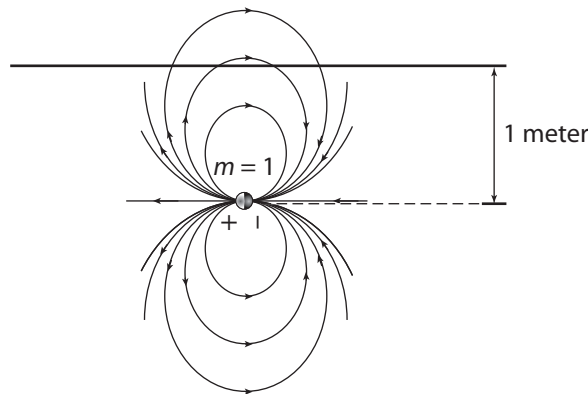


Figure 4: Horizontal magnetic dipole with moment $m = 1$ at 1 m below the surface.

3. Compute the vertical component of the magnetic field strength, B_z , at the surface of the Earth for a **vertical** magnetic dipole buried a depth z below the surface (Figure 2). Plot your result. For your axes, plot the vertical axes in terms of factors of

$$M = \frac{\mu_0 m}{4\pi z^3}. \quad (3)$$

4. Compute the vertical component of the magnetic field strength, B_z , at the surface of the Earth for a **horizontal** magnetic dipole buried a depth z below the surface (Figure 4). Plot your result as above in terms of

$$M = \frac{\mu_0 m}{4\pi z^3}. \quad (4)$$

5. **Anomaly width to depth relations**

Estimating the depth to a structure based on the width or half-width of the anomaly is a useful field tool. Using the equations for a vertical dipole, show that the depth of the dipole is approximately equal to its depth. Start by equating the solution at $x = 0$ to the solution at $x_{1/2}$.

Note, you will have to solve a 10-th order polynomial to solve this problem. You can use Wolfram Alpha online to solve for the polynomial.

For a horizontal dipole, use the peak to peak width to develop a rule for the anomaly width to burial depth. Remember to find the locations of the peaks you will have to minimize the function, i.e., $\frac{dB_z}{dx} = 0$ and solve for x .

Check your result in relation to your graphical results from Problems 1 and 2.

Question 3: Magnetic Susceptibility and Mineral Content

[10 pts.]

1. Magnetite (mt) has a susceptibility of order $\chi_{mt} \approx 1-5$. Show that the bulk susceptibility of a rock can be approximated as

$$\chi_{bulk} \approx f_{mt}\chi_{mt},$$

where f_{mt} is the volume fraction of magnetite.

2. Estimate the magnetite volume fraction required to produce bulk susceptibilities of 10^{-3} and 10^{-2} .
 3. Convert these to weight percent magnetite, assuming $\rho_{mt} = 5200 \text{ kg m}^{-3}$ and $\rho_{rock} = 2700 \text{ kg m}^{-3}$.
 4. Briefly discuss why magnetic anomalies are often dominated by small concentrations of magnetite.

Question 4: Disappearance of Marine Magnetic Anomalies

[10 pts.]

Marine magnetic anomalies weaken near continental margins.

1. Assume a linear geotherm

$$T(z) = T_s + \frac{qz}{k},$$

with $T_s = 0^\circ\text{C}$, $q = 70 \text{ mW m}^{-2}$, and $k = 2.3 \text{ W m}^{-1} \text{ K}^{-1}$. Stein & Stein (1992) developed a plate cooling model for the oceans. They suggest oceanic heat flow may be predicted as $q = 510 t^{-1/2}$ where t is the age of the seafloor in Ma and heat flow is in mW m^{-2} .

Estimate the depth to the Curie temperature of magnetite ($T_C \approx 150^\circ\text{C}$) as a function of age.

2. Determine the thickness of sediment required such that the top of the basaltic basement exceeds the Curie temperature, assuming a thermal conductivity of $1.5 \text{ W m}^{-1} \text{ K}^{-1}$.
 3. Explain how sediment accumulation, elevated temperatures, and increased depth combine to suppress marine magnetic anomalies near continental margins.